



June 8–9
Taipei, Taiwan

Asia-Pacific^{'26} Symposium

**Beyond Efficiency: Leveraging Agentic AI for Claims Experience
Analysis & Risk Resilience in Taiwan's Life Insurance**



Speaker Introduction

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Cathay Life
Insurance

Leading life insurer in Taiwan

We are committed to helping people achieve better futures with comprehensive protection and lifelong partnership.



8M+
Customers



20M+
Policies



LiYu Tsao



Senior Data Science Analyst S



Cathay Life Insurance Co., Ltd.



YuChung Wu



EXPERTISE



AI / Machine Learning
in Insurance Applications



Claims Process
and System Architecture



Agentic AI and Data Analytics



KEY HIGHLIGHTS



Built multi-agent claims pre-review
and decision-making system



Transformed claims data into
actuarial-ready information



Integrated AI with claims workflows to
improve efficiency and decision quality



國泰人壽
Cathay Life Insurance

Key Challenges in Claims Operations

Claims operations are facing structural pressure from rising risk × surging demand × workforce constraints



Escalating External Pressures

- Aging population drives sustained claim growth
- Shift of medical costs to commercial insurance
- Fraud and risk landscape becoming more complex



Accelerating Operational Burden

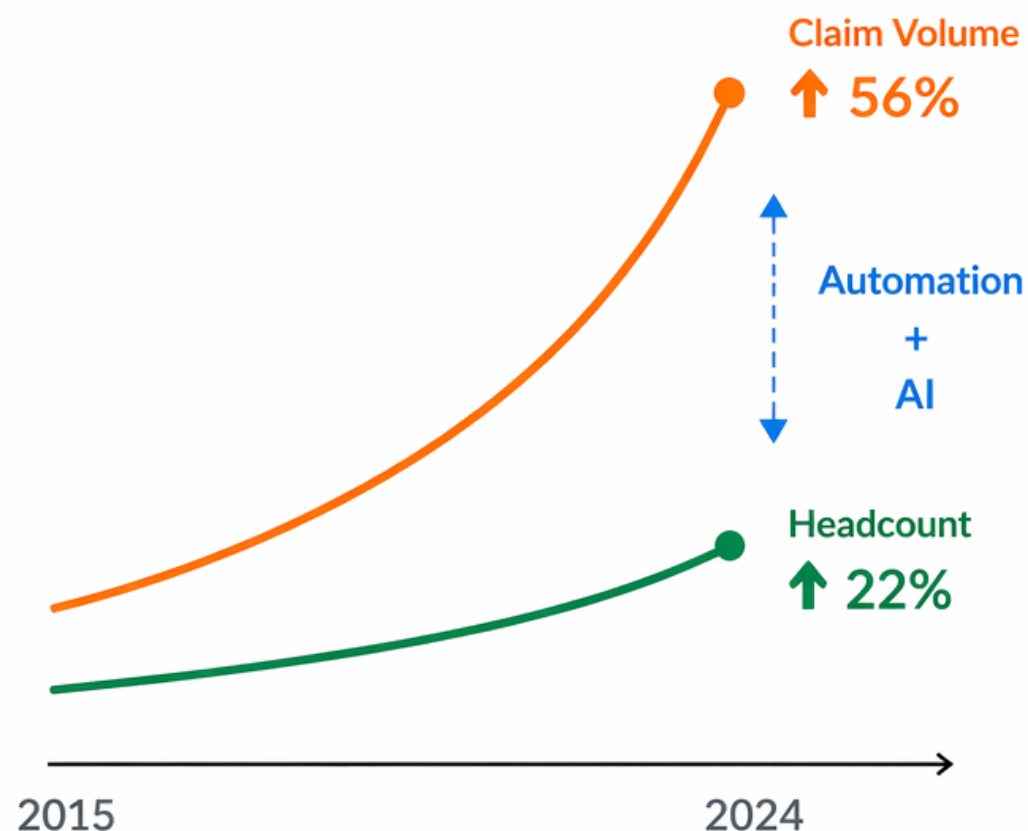
- Annual claims volume reaches 1.6 million
- Increase in high-frequency, low-value claims
- Claim volume grows +56%, while headcount only grows +22%



Talent Constraints Becoming Critical

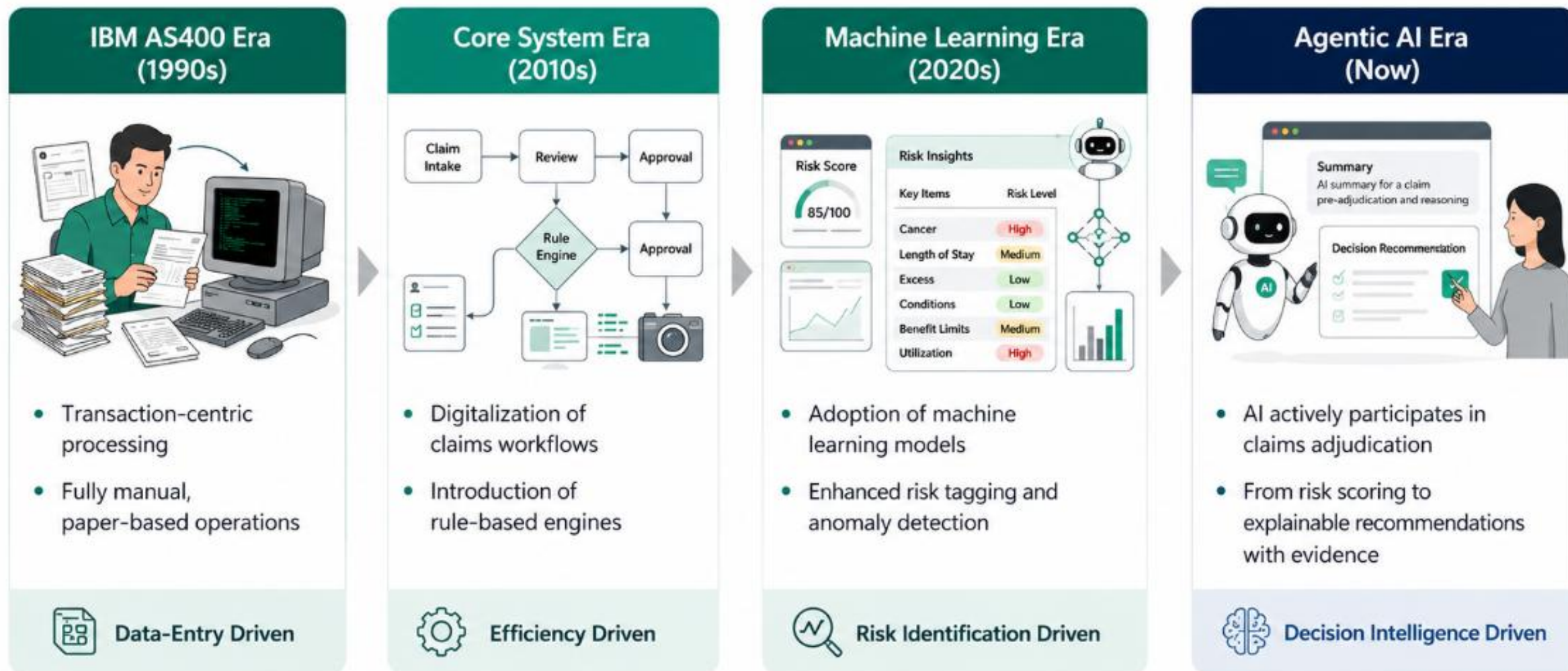
- Shortage of claims and risk control professionals
- Traditional, manual processes approaching limits

Pressure on Sustainable Growth



Evolution of Claims Adjudication

Claims processing has evolved from manual workflows to intelligent decision-making



Traditional Claims Review & Adjudication Process

An end-to-end process for claims review and adjudication, determining coverage and payment decisions



Claims Submission | Digital Application Online

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Submit claims through the mobile app, paperless application to streamline the claims process

1 Claims Application

Tap "Notify Now"



2 Upload Documents

Upload required documents
Tap the orange "+" to upload photos



3 View Claims Results

Tap "Expand Case Info"
to view payment details



4 View Application Details

Tap a case to view claim status
and remittance information



Registration & Case Creation | AI-Driven Structured Data Extraction

Automatically extract and structure medical document data using OCR and LLM models

1 Image Data Intake

Receive medical document images uploaded in the first step



Before Original Image (Medical Document)

診斷證明書					
姓名	王小明	性別	男	出生日期	1980/05/10
身分證號碼	A123456789				
病歷號碼	00012345				
診斷	右側膝關節韌帶撕裂傷				
建議	建議住院手術治療				
受傷日期	2024/05/01				
住院期間	2024/05/01 – 2024/05/05				
手術名稱	右膝關節韌帶修補手術				
開立日期	2024/05/05				
醫師姓名	陳大仁				

榮民總醫院

2 AI-Powered Structured Extraction (OCR + LLM)

OCR recognizes text and LLM understands context, then automatically extracts structured fields



1 Text Recognition
OCR extracts text from the image



2 Semantic Understanding
LLM understands the context and meaning of the content



3 Field Mapping
Map extracted content to structured fields



4 Validation & Storage
Validate fields and store structured data

3 Structured Data Output

Convert image content into structured data for claims processing workflow

After Structured Data (Extracted Fields)

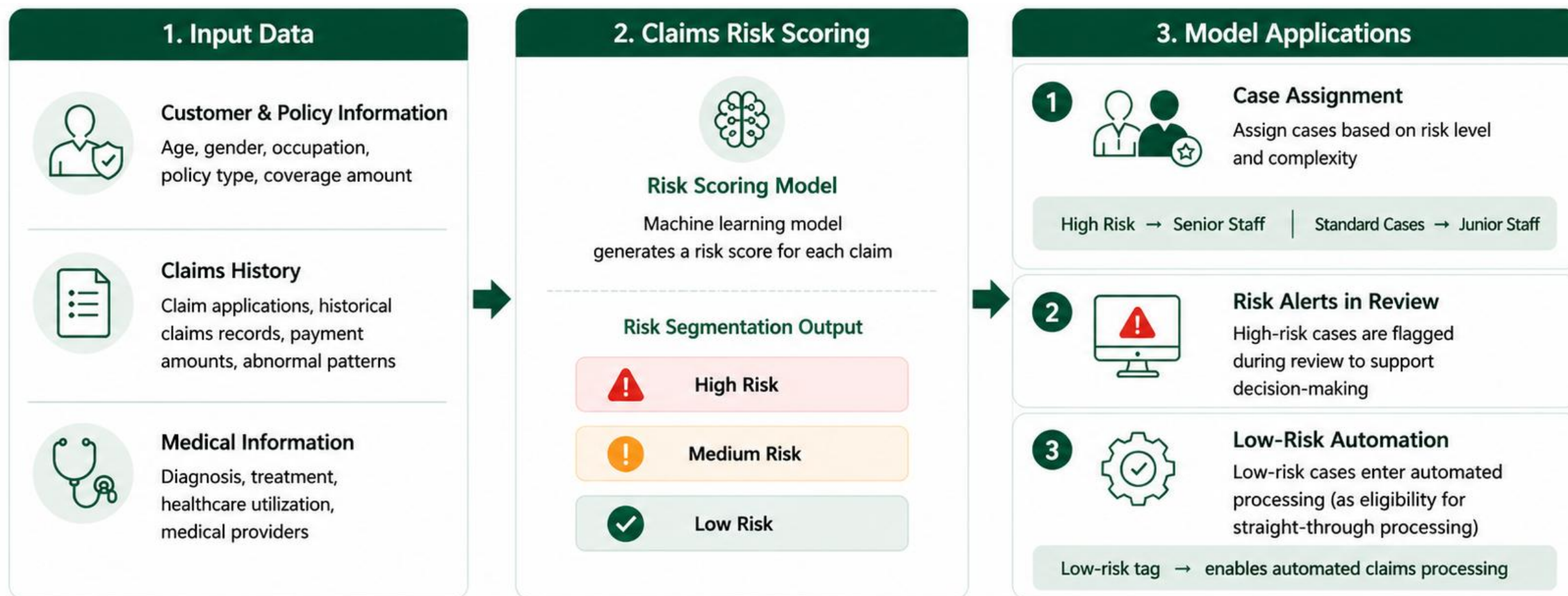
Field	Extracted Value
Name	王小明
Gender	男
Date of Birth	1980/05/10
ID Number	A123456789
Medical Record No.	00012345
Diagnosis	右側膝關節韌帶撕裂傷
Recommendation	建議住院手術治療
Injury Date	2024/05/01
Hospitalization Period	2024/05/01 – 2024/05/05
Surgery Name	右膝關節韌帶修補手術
Document Date	2024/05/05
Physician Name	陳大仁



Structured Data
Storage System

Claims Triage | Risk Scoring Model

Apply machine learning to assess claim risk, enabling case triage, risk control, and automated claims adjudication



Manual Review | Unified Smart Claims Platform

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Integrating multi-source data to enhance claims processing efficiency and quality

 Claims Information

 Review Tabs

 Risk Detection

 Quick Access

 Decision Support

國泰人壽核心系統

保單概況 | 申請資料 | 核算結果 | 受益人分配

A123456789同為精神病患者，住院期間會重疊，OO醫院主治醫師為吳OO

作業風險 (6) 挂管保戶 監理室 會招攬不
解約率 討論區 寄件 標籤

家戶(2) 疾病(1) 治療內容(1)

家戶申請浮濫
疑似家族浮濫件，被保人家戶近期已有 4 次 白內障手術理賠紀錄，建議交叉確認治療合理性再議。
同意 不同意
請輸入不同意原因

家戶申請浮濫
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同意 不同意

決策引導
互動問題 簽呈內容

Q1 保單已投保多久？
有保單未滿 2 年 皆投保 2 年以上

受編 12345678901234 OIU VIP 華人 已投保 離起碼

事故者姓名 江佩榆 事故者 ID A123456789 出生日期 79-01-23 (31歲) 國籍 中華民國

事故/索賠資料

申請類別 A	申請種類 疾病	事故原因 012-資料資料	事故原因內容 泌尿科疾病-腎炎神經
索賠類別 醫療實支, 醫療日額(燒燙傷), 津貼(生育結婚)	事故日 110-08-01	事故職等 2	

檢警資料

事故地點 台北市松山區十全路交叉路口

報案 報案日期 110-08-11 報案機關 松山分局民有派出所 報案機關電話 02-6484159#123 承辦警員 李宸睿

無記名附約

序數	類型	保單號碼	附約	對象
1	壽/意外	15424847241	資料資料資料	配偶
	員工福團	15424847241	資料資料資料	配偶

新增附約

團險資料

公司名稱	資料資料資料資料資料
事故者為員工之	資料資料資料
員工ID	資料資料資料資料資料
員工姓名	資料資料資料資料資料
員工生日	75-08-12
團險保單號碼	1234567890123
等級	資料資料資料資料資料

診斷書

快捷功能 快捷專區 輸入 理賠紀錄 特殊紀錄 索引紀錄

110-09-07 11:37 儲存管理進度

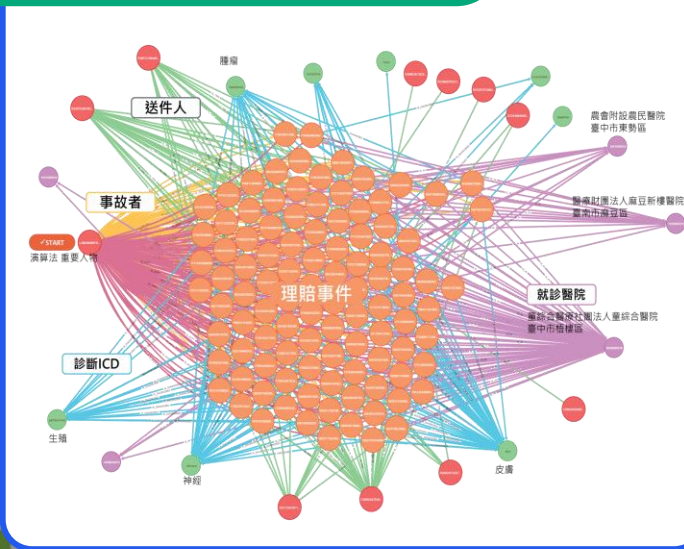
Manual Review | Embedded AI-Assisted Features

AI capabilities embedded into the review workbench to support claims adjudication

Risk tagging

作業風險 (6) 控管保戶 臨櫃常客 曾招攔不查
解約率高 討論區常客 標籤 標籤

Relationship Graph Analysis



Decision Support

互動問題

Q1 保單已投保多久？

投保 2 年以下

皆投保 2 年以上

Q2 投保兩年內保單，本次申請有無可給付項目？

有保單未滿 2 年

有可給付項目

Q3 投保兩年內保單，是否需交查確認保前疾病？

是，需確認是否保前

否，交查無實益

否，本次申請疾病已除外
或前已交查為保前

否，相同疾病已給付

簽呈內容

保單資訊

治療重點 保單皆投保兩年以上

治療重點 除外A2疾病與現症無關

審理內容

治療重點 已補全正本收據，全意依照實支計算給付

治療重點 已補全正本收據，全意依照實支計算給付

病理報告 已補全正本收據，全意依照實支計算給付

決策內容

治療重點 眼科診所門診手術，全意系列商品融通...

治療重點 白內障手術頻率合理，必要手術，可給付

簽呈預覽

無

特殊紀錄

索引紀錄

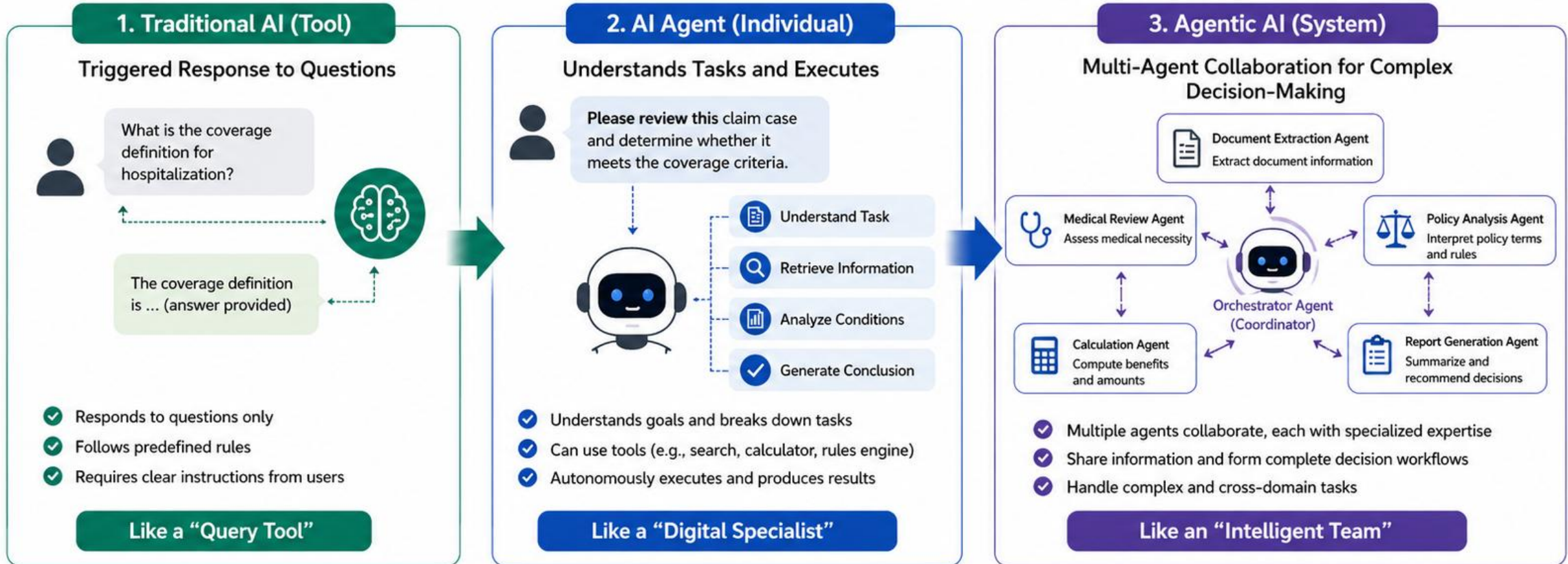
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儲存審理進度



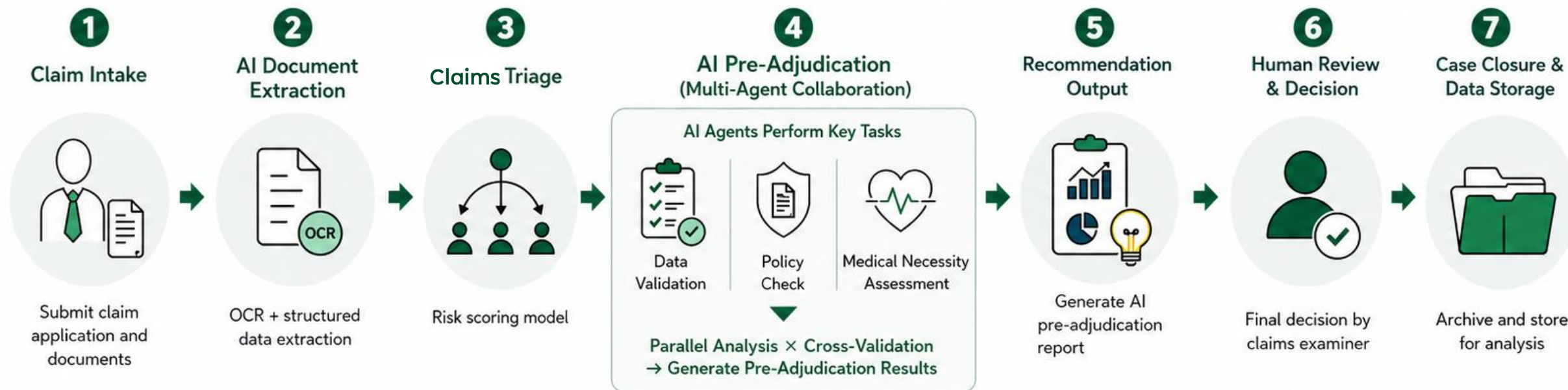
How Does AI Evolve from a Tool to a Collaborative Decision-Making System?

From reactive responses to proactive execution, and ultimately to multi-agent collaboration for complex decision-making

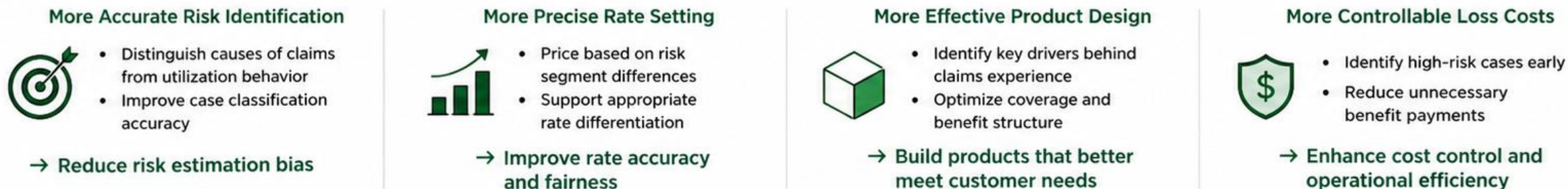


Human-in-the-Loop Claims Pre-Adjudication

Agentic AI provides analysis and decision recommendations, while human reviewers make final decisions



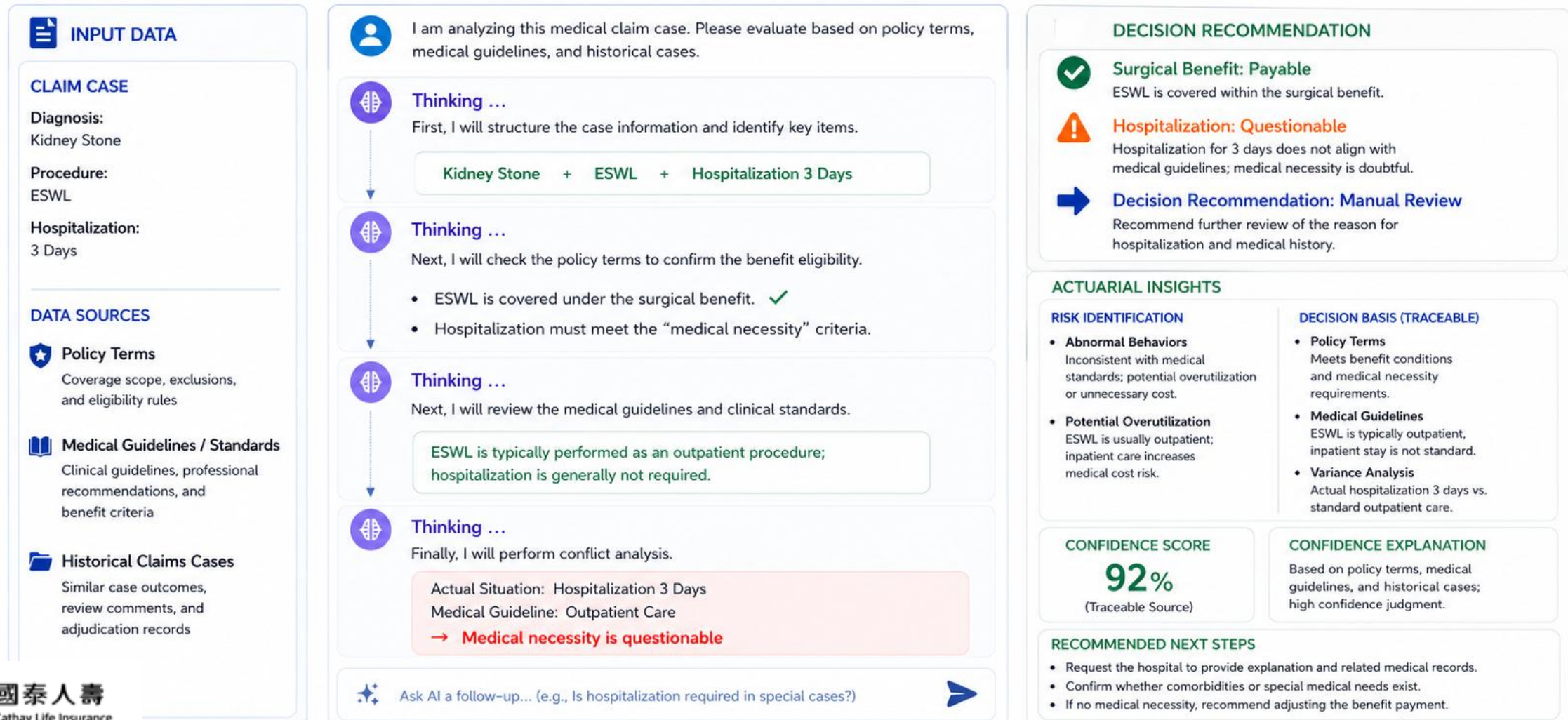
Actuarial Benefits: Enabling More Accurate Pricing and Better Product Decisions



AI Reasoning Pathway

Multi-Agent Collaboration for Explainable and Trustworthy Claims Decision-Making

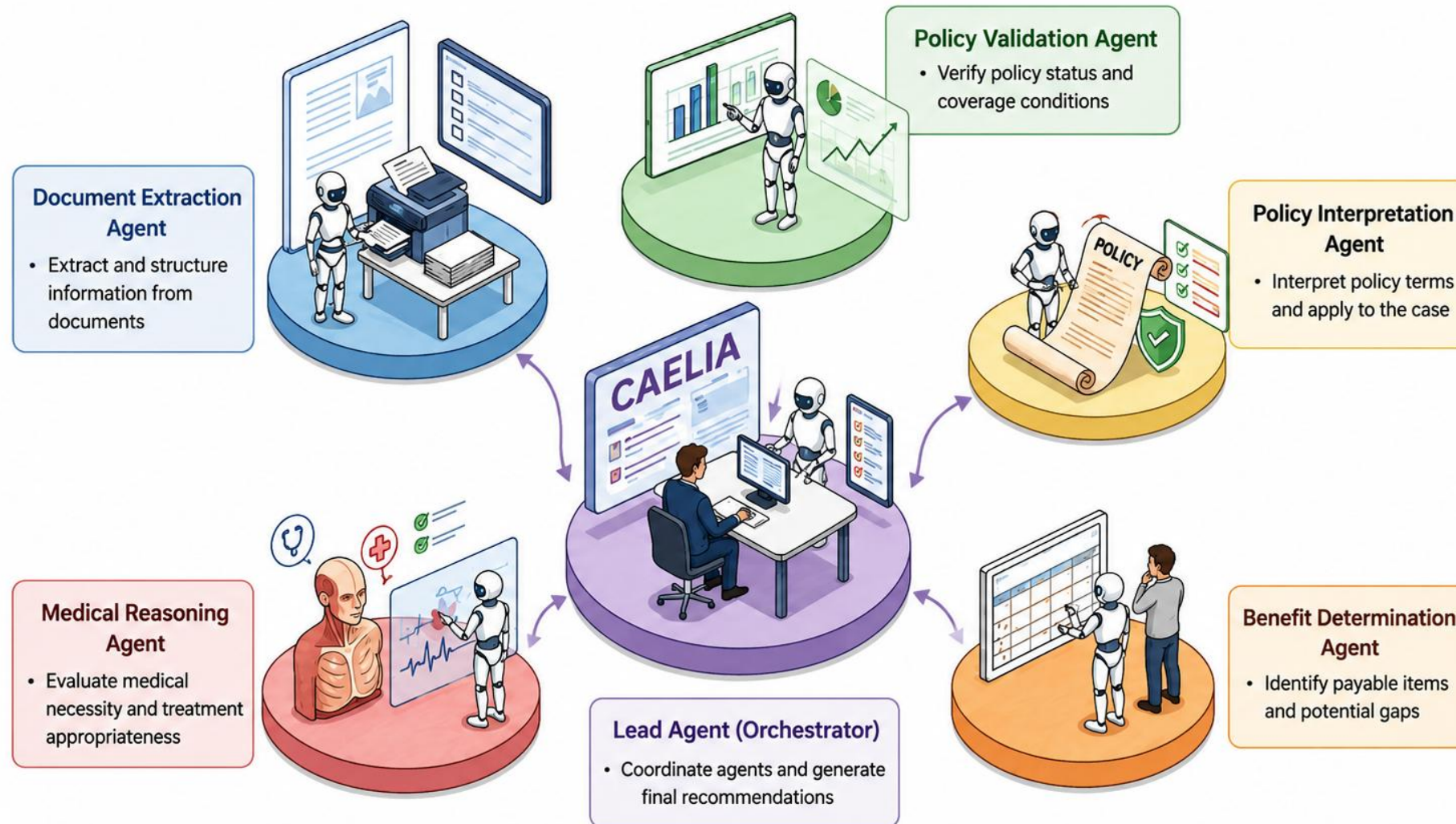
..... Orchestrated by Lead Agent (Orchestrator)



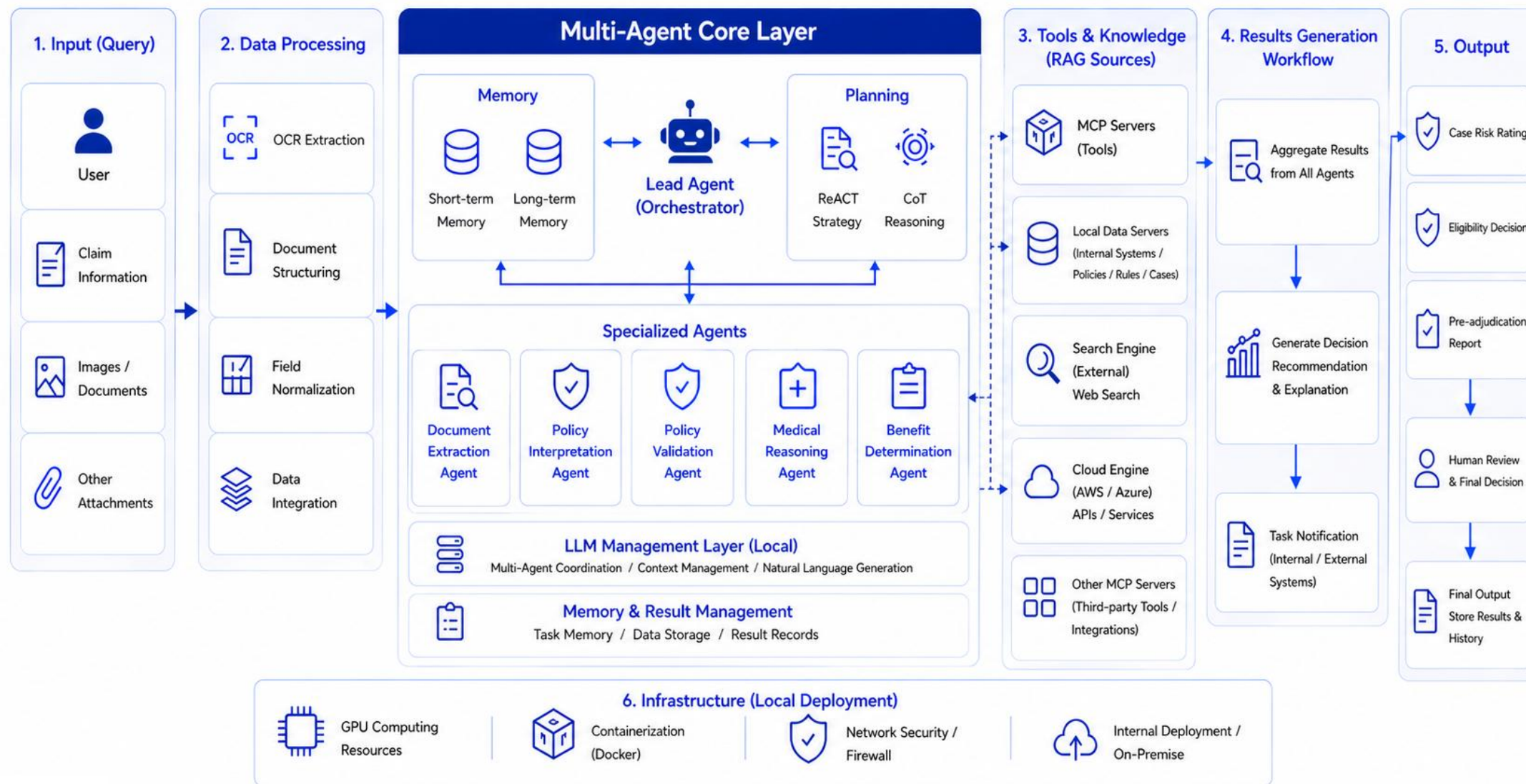
CAELIA | Cathay AI Engine for Life Insurance Adjudication

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Transforming Claims Adjudication into an Explainable and Evidence-Driven Decision Process

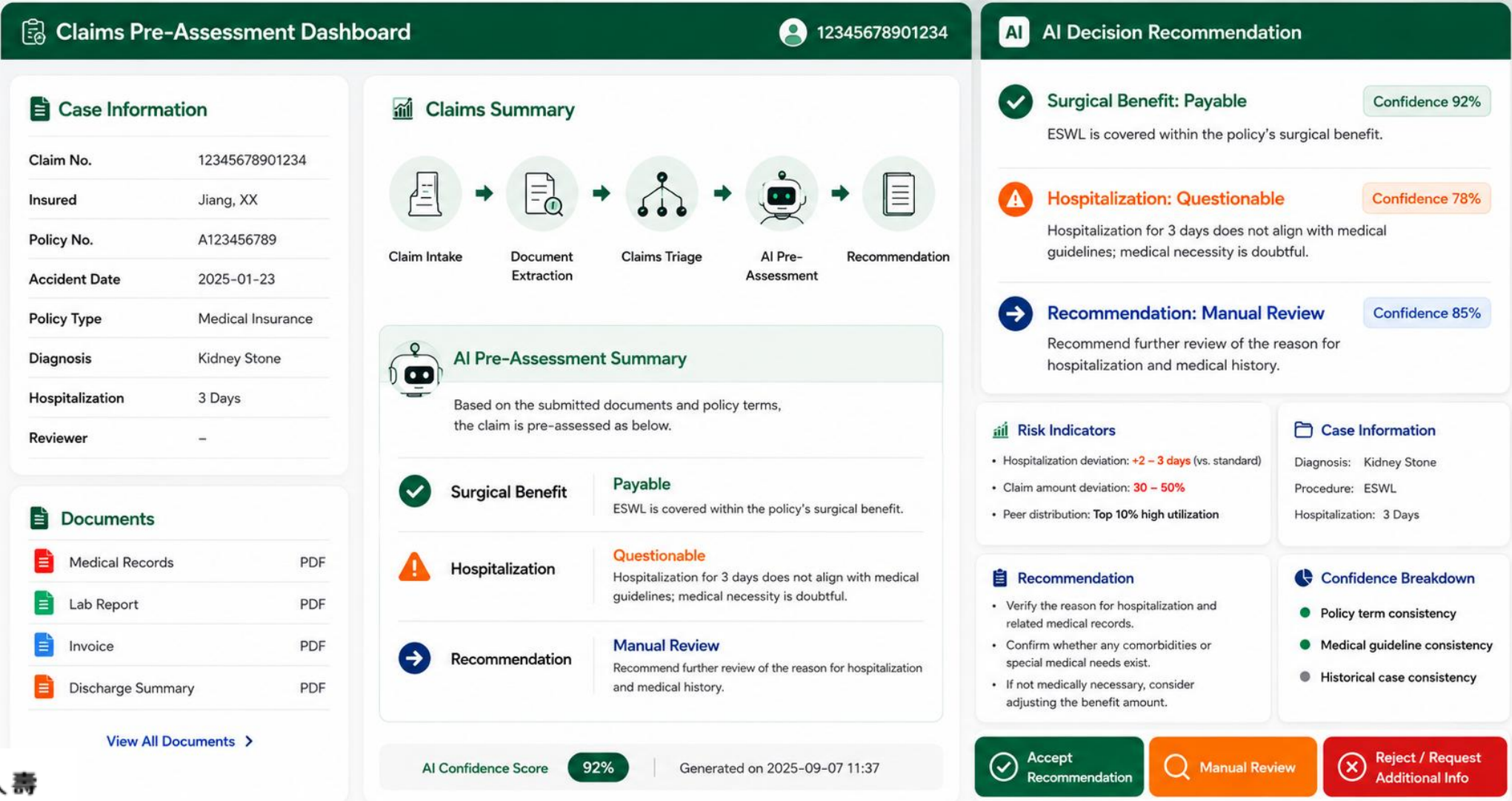


CAELIA | Multi-Agent Architecture Overview



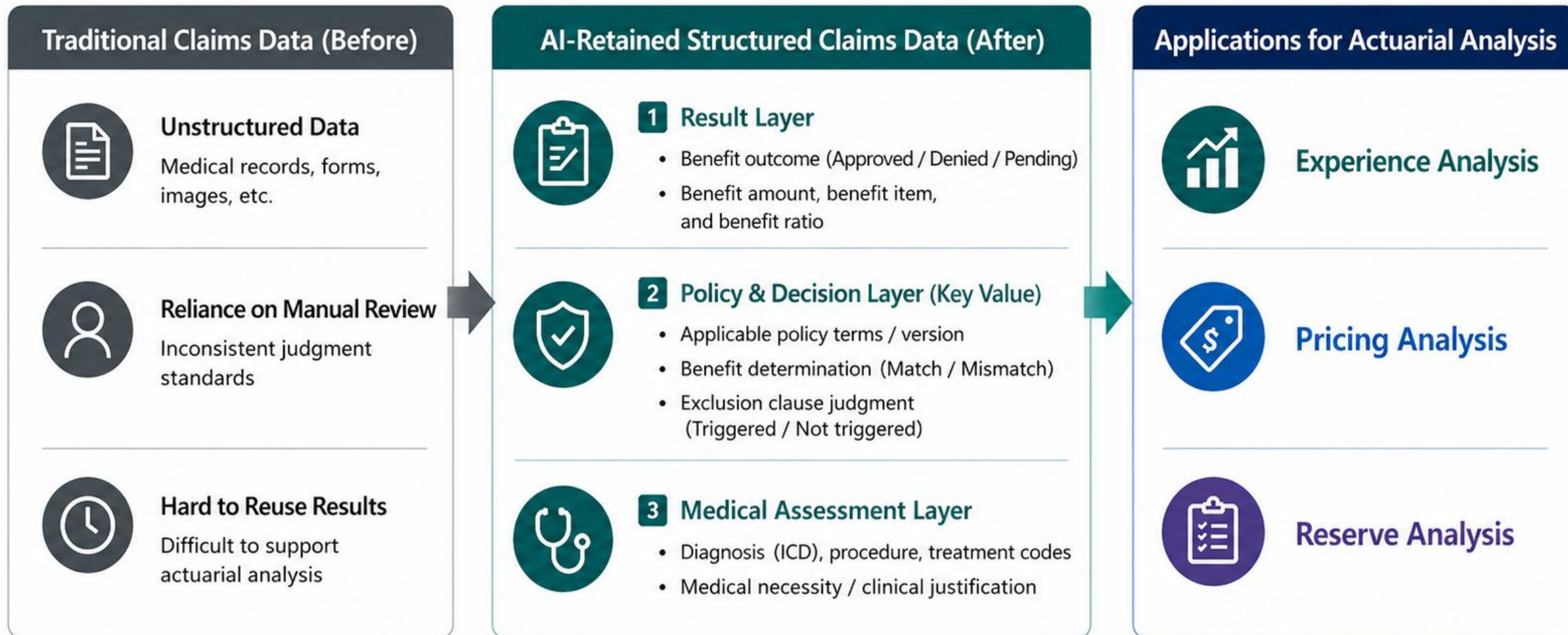
CAELIA | AI Pre-Assessment Decision Report

From raw claims data to traceable decision recommendations



From Claims Records to Actuarial-Ready Data

Transforming claims into structured data for pricing, reserving, and experience analysis



Experience Analysis Transformation: From Outcome to Cause

Improving incidence estimation through better risk segmentation and data quality

Case Context



80-year-old male



Cause of death: Fall leading to injury



Claim type: Accidental death

Before: Traditional Experience Analysis (Outcome-based)

Approach

- Based on "outcome" only (accidental death)
- All accidental death cases pooled to calculate incidence

Accidental Death Population



Limitations

- Does not distinguish external vs. internal causes (degeneration / disease)
- Heterogeneous risk pool



Impact

- Higher estimated incidence
- Systematic bias in risk estimation
- Pricing deviates from true risk



Risk overestimated
Pricing not accurate

After: AI Cause-Based Analysis

Approach

- Integrate multiple data to identify causes
- Analyze "disease risk" and "utilization behavior"
- Apply different strategies by risk source

Cause Classification



External cause
(pure accidental)



Internal cause
(degeneration / disease)

Output

- Flag "disputed cases"
(non-pure accidental)



Disputed Cases
(require further review)

Model Application

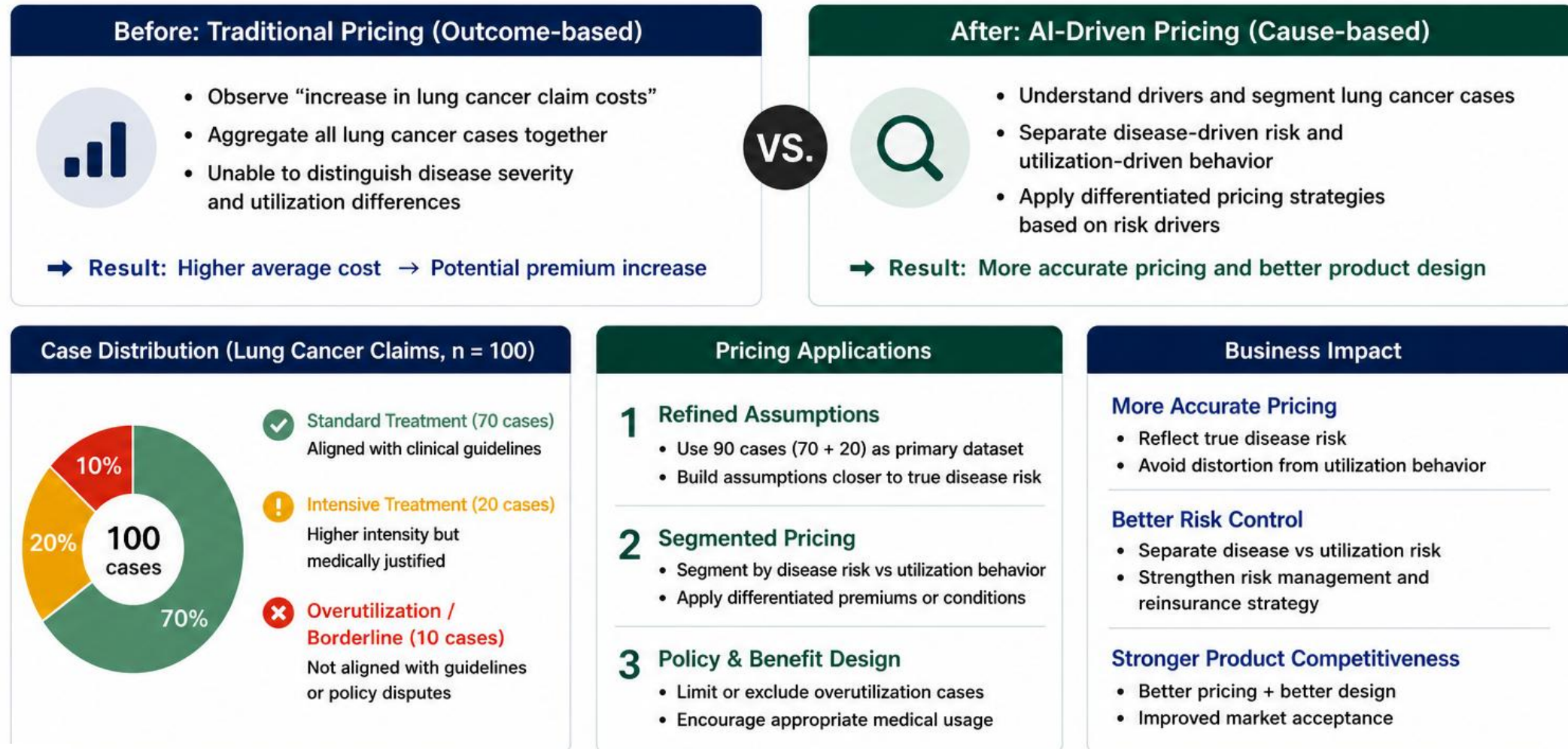
- Exclude or reweight (downweight) disputes
- Build a cleaner and more homogeneous claim population

Analysis Population (Cleaner)



Pricing Application Extension: From Cause Analysis to Risk-Based Pricing

Transforming risk decomposition into differentiated pricing strategies



CAELIA | Knowledge Foundation

Structured and governed knowledge for explainable AI decision-making

1 Three Core Knowledge Sources



Claims Operations Policies

Key internal guidelines and review focus areas



Product Terms & Rules

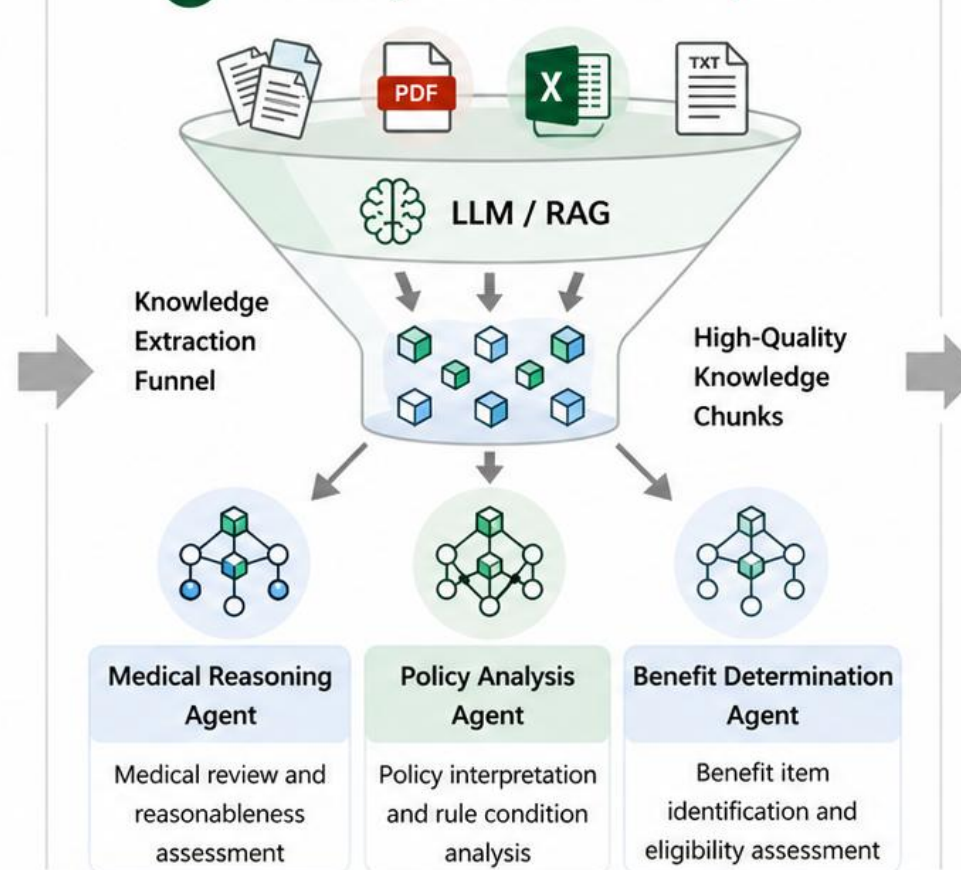
Policy terms, benefit conditions, and limitations



Medical Knowledge & Disease Library

Medical guidelines, disease classification, and treatment protocols

2 Knowledge Transformation Pipeline



3 Tech Governance & Monitoring



Traceability

Citable sources for every conclusion, supporting auditability and traceability



Data Privacy & Bias Mitigation

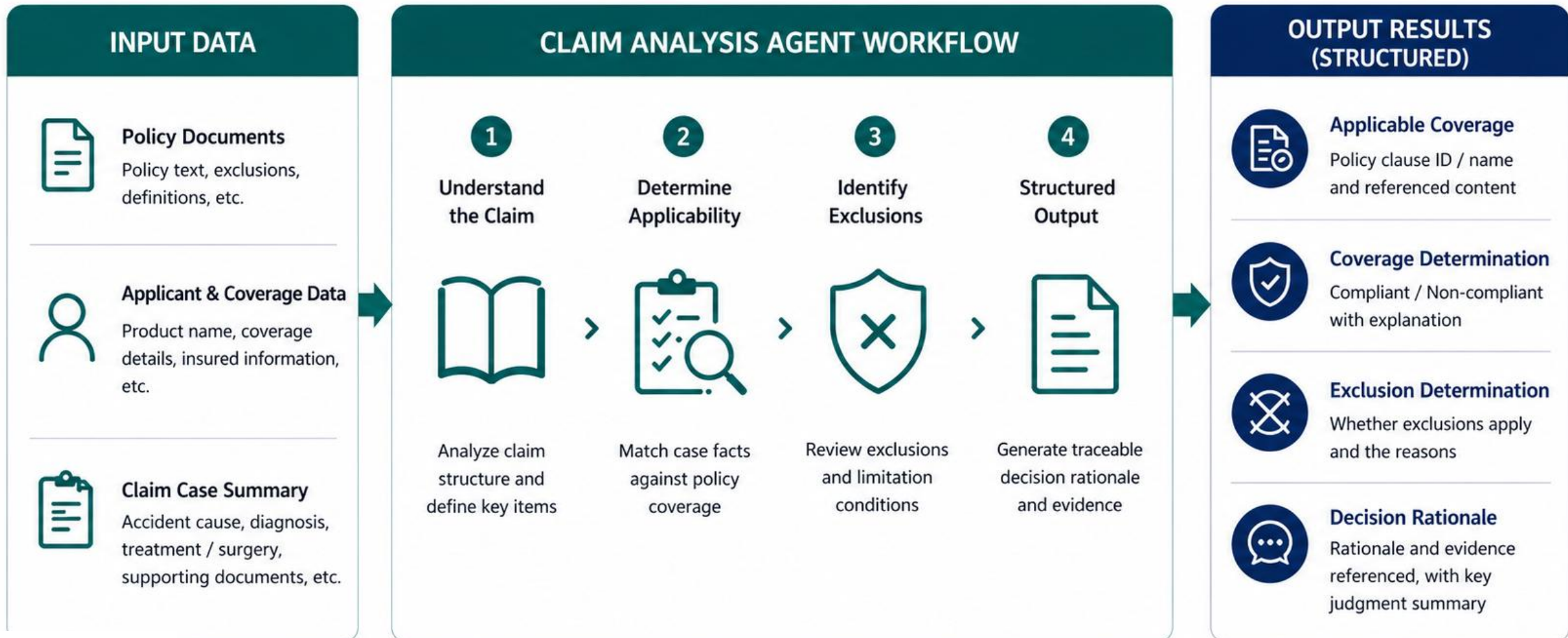
Minimize data usage and ensure compliance for secure and responsible AI



Structured, governed knowledge enables consistent, trustworthy, and explainable decisions.

CAELIA | Policy Interpretation

Analyze policy terms to determine coverage, exclusions, and decision rationale



CAELIA | AI Pre-Adjudication Benefits:

Operations × Risk Control × Governance × Actuarial

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Interpretable and governable AI transforms claims data into the analytical foundation for actuarial insights



Service FTEs

300 ↑ FTEs

Across the Claims Department



Save

30%

of Review Time
per Case



Society of
Actuaries®